

Spam Movie Reviews Detection through Supervised Learning

Chun Sang Au Yong and Wing Yan Wong

Department of Computer Science

The Chinese University of Hong Kong

{1155143521, 1155143285}@link.cuhk.edu.hk

Abstract – Identification of spam reviews is an important research topic with many applications. However, existing research has focused on commercial goods but not products from the entertainment industry. Recognising the influences of spam reviews on the movie sector, which is enormous in market value, this project aims to build a dataset from scratch by labelling reviews online as spam or genuine, and then identify useful features and effective models to detect spam movie reviews, enabling more sophisticated spam review filters which can combat the omni-presence of spam reviews online.

Keywords – spam reviews; dataset construction; methodologies; natural language processing; machine learning

1. Introduction

In recent years, especially amid a pandemic which promotes indoor entertainment, there has been a surge in demand for movies. The growth of movie industry has, however, led to the emergence of spam reviews. Paid posters, hired by movie distributors who want to promote by word of mouth, or attack movies made by competitors to achieve market dominance, will post false opinions in an attempt to manipulate audience's opinion. Positive opinions provide considerable financial gain, whereas negative opinions often cause box office losses [1, 2]. Flooded with spam reviews, review platforms fail to provide trustworthy guidance to moviegoers. In China, this phenomenon is especially critical as paid posters, also known as the Internet Water Army, are quite common across websites [3].

Previous research on commercial product spam review detection may be fruitful, but few focuses primarily on movie reviews which have unique word distribution and thus unique features. In the scope of movie review, only a handful of research has studied reviews in Chinese. Besides, output from previous research cannot be applied to combat the prevalence of spam movie reviews because methods proposed are also not effective against spam reviews in the real-life [4].

Our project revolves around spam reviews on a Chinese movie review site, namely Douban Movie (www.movie.douban.com). Notorious for pervasive spam reviews, this site has received widespread

media coverage [5, 6]. Scraping and labelling reviews on the platform, we obtain a dataset on which we can engineer features, conduct exploratory data analysis, and train models.

By introducing AI into spam movie review detection, we hope to minimise the presence of spam opinions so that confidence in review platforms can be restored. This is of paramount importance to both consumers and producers. No longer clouded by false opinions, moviegoers can select movies with reference to the genuine reviews which serve as an honest guide providing useful recommendations. Film production companies also benefit from the decline in spam reviews as they can better assess the reception of movies by analytics, which is more effective without the noise created by spam reviews.

This project therefore provides significant contribution in the field of fake review detection. First of all, a labelled dataset containing 1600 realistic reviews is created, enabling further research in fake review detection using supervised models. Results from the EDA of features can also provide insights into the usefulness of features, so that later research can use this as a guiding principle to avoid wasting resource on extracting features that are futile to spam review detection. Comparing the performance of various models, this project can help create more sophisticated spam filters in movie review websites by identifying the most effective algorithms in a real-life setting.

2. Related Work

Reviews in other sectors were formerly researched. Both semantic and behavioural features which could be useful in detecting spam reviews were proposed [7]. Although previous experiments reported better accuracy using behavioural features on real-life commercial spam reviews [4], we decided to attach greater emphasis on semantics in this project considering that the text is core to each review, while behavioural features may vary from site to site and are often unavailable due to site restrictions. Although previous studies showed that in addition to supervised learning [8], unsupervised learning [9] and graph learning [10] can also be used, we chose supervised learning for its simplicity and effectiveness.

Previous researchers also proposed several ways to produce a dataset. Some proposed generating spam reviews using a text generation AI,

namely GPT-2, to produce spam reviews [11]. However, they overlooked the lack of evidence showing that paid posters actually use text generation AI, so generated reviews may not reflect the actual spam reviews. On the other hand, since it can hardly be said that the model used is the best available across the market, one could easily doubt the performance of algorithms by claiming that the text generation AI is incompetent, or simply not state-of-the-art.

There were also researchers who attempted to build datasets by merging deceptive opinions generated in crowdsourcing services such as Amazon Mechanical Turk (AMT), and truthful opinions obtained online [12]. Despite clearing the aforementioned doubts on text generation AI, this is still not a plausible approach. Although we can ensure that the generated reviews are deceptive opinions, no guarantee can be made regarding the ‘truthfulness’ of the reviews obtained online, especially amid the omnipresence of spam reviews online. Besides, reviews written by anonymous online workers (Turkers) may not be representative of real-life spam reviews because of the difference in word distribution [4]. Thus, algorithms achieving a high accuracy in detecting crowdsourced spam reviews may not be effective in real-life settings. The cost of research will also skyrocket because each review written generated by AMT costs one US dollar.

3. Methodology

Fig. 1 presents the framework of the proposed methodology. It consists of 7 steps.

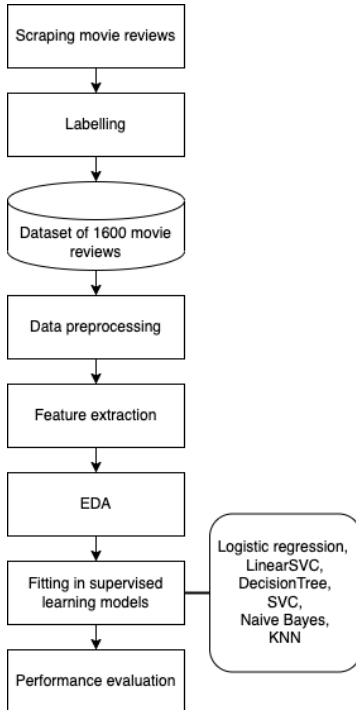


Fig. 1. The proposed methodology.

We commenced with extracting movie reviews from Douban Movie using the Python library BeautifulSoup, which transforms HTML documents into a tree of Python object. A web scrape program was written to scrape information of reviews of any specified movie by providing its unique identifier on the website. We scraped from a list of movies, both Chinese and Western, which are suspected to have a high proportion of spam reviews.

Acknowledging the mix of spam and genuine reviews online, we turned to human labelling. Upholding the principle of Occam’s razor, this simple yet intuitive approach is very desirable in creating a dataset provided that two conditions are satisfied: (1) spam reviews are abundant online and (2) reviews are correctly labelled. In the setting of previous research, (1) was often violated because they assumed that most reviews online are genuine. For example, in [12], reviews online were chosen to constitute the genuine part of the dataset. It would therefore not be sensible for them to collect spam reviews by scraping the web. Regarding (2), while the accuracy of human judge alone may be questioned, we invited multiple experimenters from varying background who read and write reviews regularly. We are confident that their experience in the field will attribute to accurate labelling. Moreover, we will provide them with prompts derived from previous research [13, 14] and common knowledge which outlines the major characteristics of spam reviews to aid their judgement. On top of that, the authenticity of each review is decided by the majority vote, so that results are more reliable as individual misjudgement can be averaged out.

Upon labelling the reviews by human effort, we obtained a labelled dataset. Containing punctuation and stop words, review texts are in general messy. We therefore cleaned the review text by stripping punctuation marks (, . ? ! : ;) and removing stop words. To further enforce a standard on the text, we resolved the mixture of Chinese character set by converting all characters to simplified Chinese which is more compatible with the NLP libraries we use. Tokenisation and sentence segmentation were also adopted to structure the text data.

We could therefore proceed with the cleaned data to feature engineering. Much emphasis is attached to this section because previous research focused mostly on tuning hyperparameters and training advanced models in attempt to achieve high accuracy, overlooking the importance of generating efficacious features specific to the problem in hand. Distinguishing between genuine and spam reviews,

a good feature should show a notable difference between classes, thereby serving as crucial guidance for training the classifier. They should also be interpretable in a sense that their effectiveness can be justified. In this study, an exposition of TF-IDF, sentiment score, count of characters and features specific to movie reviews will be conducted to probe into their usefulness and the reason behind.

Term frequency-Inverse document frequency (TF-IDF) is a useful statistical approach to scale how significant a word is to a document in the dataset [15, 16]. TF-IDF is the product between TF and IDF: the former measures the frequency of a word in a document, while the latter measures the occurrence of a word in the entire document set. This statistic works by increasing proportionally to the number of times a word appears in a document yet offset by the number of documents that contain the word. As a result, words that are common in every document are ranked low, while words specific to a single document are ranked high. TF-IDF can therefore encode words by their respective relevance so that classification algorithms can understand them.

Sentiment score also plays a significant role in detecting spam reviews as previous research showed that extreme sentiment is usually present in spam reviews [8]. We adopted the lexicon-based sentiment analyser in SnowNLP, which is a pre-trained naïve Bayes classifier based on a defined dictionary. The model goes through words in reviews to count the occurrence of words grouped by their sentiment polarity to produce a probability of the text being positive, ranged from 0 to 1.

Instead of just abiding by the standard features in NLP, a range of features specific to the sector of movie review was engineered. We hypothesised the use of emoji, English and numbers, as well as the length of review before and after cleaning could be useful in differentiating spam and genuine reviews. In many other literatures, the review text was simply cleaned thoroughly, retaining few traits that could be useful for classification. This project, in stark contrast, preserve these subtle attributes by counting the characters removed. For example, while there are no characters from other languages present in the English dataset created by [12], we counted the number of these characters as a separate feature before removing them from the text, allowing us to better capture the semantic complexity of reviews.

We then performed exploratory data analysis on the extracted features to observe their distribution in spam and genuine reviews respectively. Plots of features grouping by class were created to inspect

the difference in distribution between spam and genuine reviews. Two approaches are available for determining whether the distributions are different or not: (1) direct observation and (2) statistical tests. While a non-parametric two-sample test such as maximum mean discrepancy (MMD) [17] excels at drawing quantifiable conclusions at various confidence levels, we decided to adopt the former because it is straightforward and we can easily conclude what are the differences in addition to whether there are any differences.

Selecting the features, we then proceed to modelling. The goal here is not to fine tune a specific model for high performance, but to try a variety of models, including naïve Bayes, k nearest neighbours (KNN), logistic regression, support vector machines with linear and RBF kernel and decision tree, to see which are more feasible and effective in spam review detection with the engineered features, as they have their own individual advantages when it comes to implementation.

The problem in hand, fake review detection, is a standard binary classification problem which involves assigning a binary label, 0 for genuine or 1 for spam, to each review. Before training the models, the dataset is shuffled and split into training set and testing set, where the former contains 80% of the reviews for training the model while the latter contains 20% reviews for testing how well the model perform on data unseen in training.

Naïve Bayes is a conditional probability model. Given a review, it can find the probability that it belongs to fake reviews or genuine reviews. It holds the naïve assumption that the value of a particular feature is independent of the value of any other feature, given the class variable. We chose to experiment with this model because it works with even a small number of training data. Variants of Naïve Bayes exist for different applications. The classic Gaussian Naïve Bayes suffer from presence of too many independent variables created from TF-IDF, so the conditional probability of these features almost perfectly fits the data. We will instead use multinomial Naïve Bayes which is a classic variant used in text classification.

KNN is a non-parametric classifier. Given a review with unknown class, it identifies the most similar K reviews from the training examples. In other words, there is no model because mapping is done during prediction, leading to fast training but slow prediction. Yet, this nature of lazy learning is useful for a practical spam review filter since model update is not necessary when new reviews arrive.

Logistic regression is the application of linear regression in classification. In this model, logit transformation is applied on the linear function to fit the probability of the classification problem. Set at 0.5, the probability threshold will draw a decision boundary, classifying all reviews whose probability exceeds 0.5 to be fake reviews. The major edge of using logistic regression comes from characteristics of the derivative of logistic sigmoid function. On one hand, the derivative changes significantly around 0 so labels can be distinguished effectively; on the other hand, it changes mildly for extrema so it is robust to outliers. In addition, the model is also easily interpretable as the weights from regression can provide insight into the relative importance of the features, provided that the model performs well.

SVM is a linear classifier which aims to find a separating hyperplane with the thickest margin as the decision boundary. Having a larger distance to the closest point, the boundary is more noise tolerant, leading to a more robust classifier. Kernel function is used to increase dimension so that data points become more linearly separable. Two kernel functions, namely RBF kernel and linear kernel, are used in this experiment. The former should typically achieve better performance since it is nonlinear, while the latter should be faster in training.

Decision tree utilises the structure of binary tree to help classification. It splits the reviews based on attribute testing that optimise Gini Index, the impurity measure. The splitting repeats until Gini Index no longer improves, or when the tree reaches maximum depth to prevent overfitting. The label of an incoming review can be predicted once it traverses the tree to reach a leaf according to the condition in each node.

While evaluating the models, we not only looked at accuracy, but also F1 score and area under ROC curve (AUC). Although our dataset is class balanced, F1 score and ROC score may still serve as useful reference in our study. We also investigated if the model is practical when used in a real-world spam filter considering its complexity and limitation, which are often neglected in other literatures as they race for high accuracy.

4 Results and Discussion

4.1 Data Collection

Limited by Douban restrictions, we randomly fetched 600 reviews from each movie on the curated list. Each review record contains user ID, review text, movie score, time created and the number of users who find the review useful. Taking reference from

related work, an ideal dataset should contain 1600 reviews where half of them is spam. Therefore, we repeatedly scraped reviews from the list of movies until we obtain 800 spam reviews.

Five independent experimenters were provided with a set of predefined rules in the process of labelling. The set of prompts are:

- Does this review include information about the movie, such as the name of movie, cast, characters, and plot? A paid poster typically does not watch the whole movie before writing the spam review.
- Does this review attempt to deter or encourage you to watch the movie? A paid poster typically attempts to influence users for the box office.
- Does this review express personal sentiment towards the movie? A paid poster typically only expresses general attitudes but not personal emotions.
- Is this review written right after, or even ahead of the movie release? A paid poster typically posts as early as possible to maximise the influence on box office.
- Is this review related to the movie? Irrelevant reviews are means to simply affect the rating with minimal effort.
- Is the user account suspended by Douban Movie? A paid poster account is likely to be suspended due to abnormal operations.

Taking the majority vote, we eventually obtained 814 spam reviews out of 5023 scraped reviews. Merging the spam reviews with an equal number of genuine reviews, we created the final class-balanced dataset with a total of 1600 rows, in which 800 are genuine and 800 are spam. The overall statistic of the dataset is shown in Table I.

Table I
DATASET STATISTIC

Movie	Total	Spam
Moon Man (獨行月球)	562	88
Too Cool to Kill (這個殺手不太冷靜)	599	85
Detective Chinatown 3 (唐人街探案 3)	556	98
Upcoming Summer (盛夏未來)	534	72
A Little Red Flower (送你一朵小红花)	593	106
One Week Friends (一周的朋友)	534	151
Coffee or Tea (一點就到家)	535	83
Mulan	510	86
Soul	600	45
Total	5023	814

4.4 Feature Analysis

The usefulness of features was evaluated by exploratory data analysis (EDA) using violin plots which can effectively shows the mean, range, and distribution of the features grouping by class, where 0 stands for genuine while 1 stands for spam reviews.

Fig. 2 and Fig. 3 shows the count of Chinese characters before and after stop word removal. The range of number of characters in genuine reviews are notably larger as it spans from 25 to more than 250 characters, while spam reviews concentrate in the range between 0 to 50 characters. On the other hand, the median in spam reviews is almost half of that of genuine reviews. From the kernel density estimation, it can be observed that spam reviews have a much higher probability of being short in length. This can be attributed to the fact that paid posters are paid on the number of reviews written instead of the length of the review [6]. As a result, numerous short reviews will help maximise income when compared to a small number of long reviews. On the other hand, genuine users on the site tend to write longer as moviegoers tend to share the takeaway from the movie, criticisms on plot and acting, favourite part after the show, let alone reviewers who are even more motivated and passionate to share their feelings online. A long review would therefore be necessary to cover the large variety of topics. Yet, the similarity in both figures also suggest that these 2 features are highly dependent. As a result, we will only keep the count of Chinese characters before stop word removal in training.

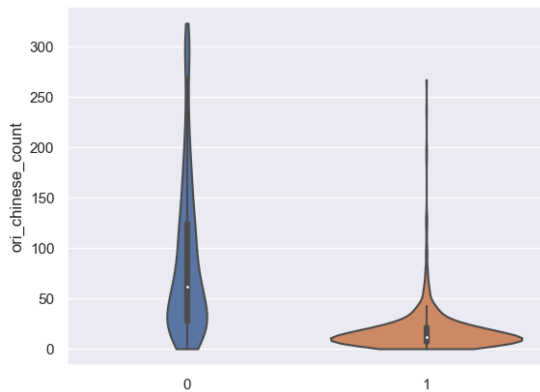


Fig. 2. Count of Chinese characters before cleaning.

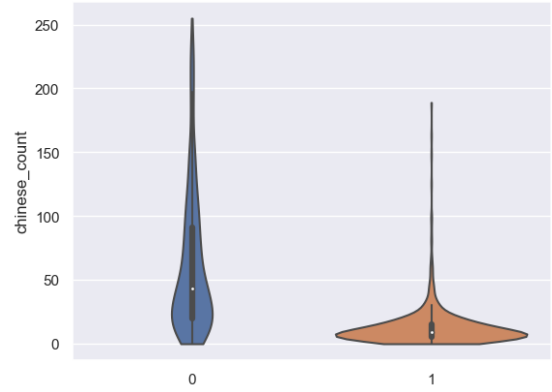


Fig. 3. Count of Chinese characters after cleaning.

Fig. 4 and Fig. 5 shows the count of English words and numbers. Although the median of both classes sits at 0, showing that English and numbers are lesser seen in Douban Movie where users mostly communicate in Chinese, alternative characters are still sometimes used in reviews, especially the genuine ones. The width of the kernel density around 0 for spam reviews is significantly wider than that of genuine reviews, suggesting that spam reviewers almost never use English and numbers while genuine users utilise these characters from time to time, albeit seldom. This may be due to the fact that rich vocabulary is essential to commenting on movies as it is sometimes more natural to use jargons and proper nouns in English, as well as describing years, age, and budget in numbers. Paid posters, on the other hand, will avoid using this vocabulary in order to blend into the general public. Simple Chinese words are also more effective in influencing others because not all users on the website are proficient in English.

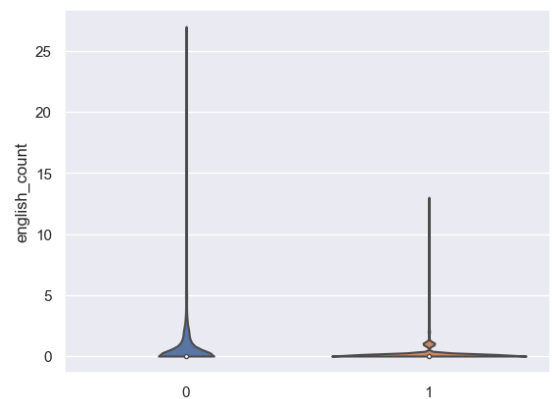


Fig. 4. Count of English words.

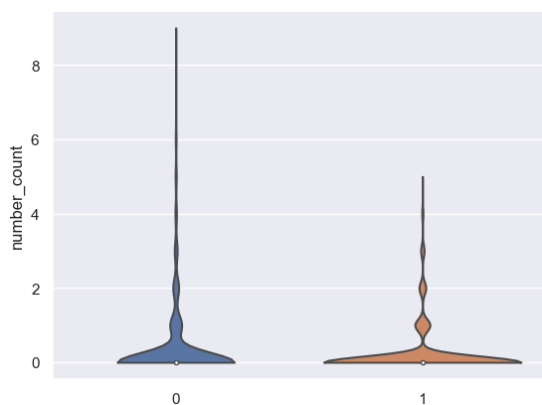


Fig. 5. Count of numbers.

Fig. 6 shows the count of emojis used in review. The distributions in both classes are remarkably similar with identical range and median. We initially thought that the use of emojis would be indicative to genuine reviews because moviegoers may rely on emojis to passionately express their feelings after watching the movie. Yet, according to the figure, the use of emojis is simply too infrequent for this phenomenon to be observed. Since the count of emojis does not help in differentiating spam reviews from genuine reviews, this feature will be removed from the modelling.

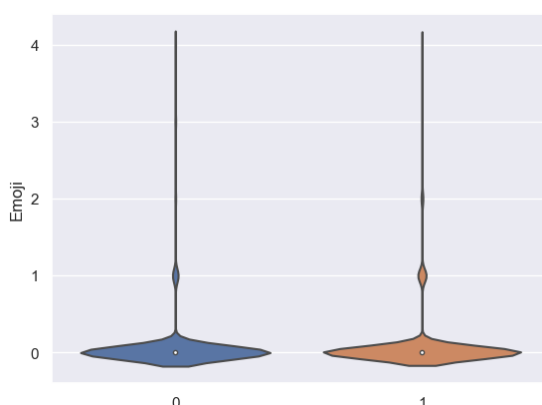


Fig. 6. Count of emojis.

Fig. 7 shows the number of appearances of the name of movie in the review. Although the median of both classes sits at 0, from the kernel density it can be seen that genuine reviews tend to mention the name of movie more while spam reviews have virtually no probability of mentioning the name of movie. Inspecting individual reviews, we found that some genuine reviews have an introductory tone which resembles that of a film criticism. It looks as if the user is a movie critic who copied from his own blogging site to Douban Movie. Mention of the movie name is essential, sometimes inevitable, in film criticism where the writer attempts to refer to specifically to the movie, as in this example of genuine review in our dataset:

其实蛮遗憾的是，这几年把校园青春片拍得好的，都是香港和台湾的导演，如果说《少年的你》是 BG 的狂欢，那《盛夏未来》就是 BL 的暗喜，几乎是内地院线电影可以做到极致了，所以加一星鼓励。吴磊饰演的郑宇星，在我心中是可以媲美刘小北一样的角色，几乎满足了所有我对少年的幻想，当然张子枫不是周冬雨，导演边拍脑海里想的应该是桂纶镁。看完出来朋友说作为一个 gay 男主的音乐审美也太烂了，我倒觉得唯一让我出戏的是打碟的地方，作为一个迷恋电子乐的 gay，梦想应该不是在 SPACE 打碟....吧？

The nature of film criticism therefore contributes to the larger number of movie name count.

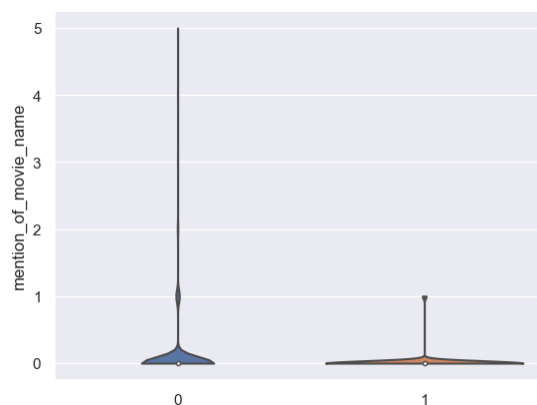


Fig. 7. Count of movie names mentioned.

Fig. 8 shows the distribution of sentiment score in the review. While the medians are about the same, the range of sentiment in spam reviews almost doubles that of genuine reviews, leading to a larger variance. This polarisation is understandable from the perspective of paid posters because strong wordings are more effective in influencing the decision of others, so the sentiment score tends to pull apart from the median.

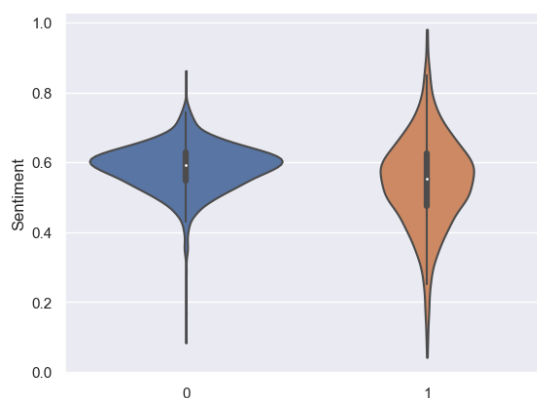


Fig. 8. Sentiment score of review.

Fig. 9 shows the distribution of review score which comes with the review text. Similar to the sentiment score, polarisation is observed in spam reviews but not genuine reviews despite identical medians. In a lot of cases, spam reviewers give the highest or the lowest ratings available not just because of the psychological influence on others, but

an extreme rating is also useful in exerting maximum influence on the overall movie score. In fact, looking at the inter-quartile range (IQR) in spam reviews, a tendency towards low rating is also observed as the IQR spans from 1 to 4, showing that paid posters on Douban Movie are mostly hired to depreciate movies.

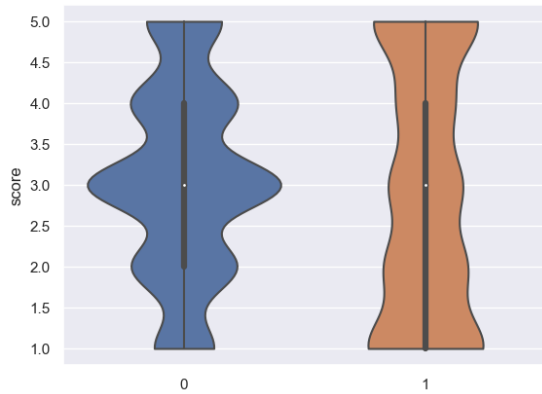


Fig. 9. Review score.

Fig. 10 shows the distribution of the posting time of the review. The kernel density for genuine review is significantly wider at night than in the afternoon, while spam reviews show a much smaller discrepancy between these 2 time periods. This is comprehensible assuming that people tend to post reviews right after watching the movie. Many have to work on weekdays, so watching a movie in the afternoon is usually impossible. Paid posters, on the other hand, can post anytime because they do not have to watch the movie before posting. Instead, they mass produce spam reviews during their working hours and posting them immediately, leading to a wider kernel density in the afternoon when they work.

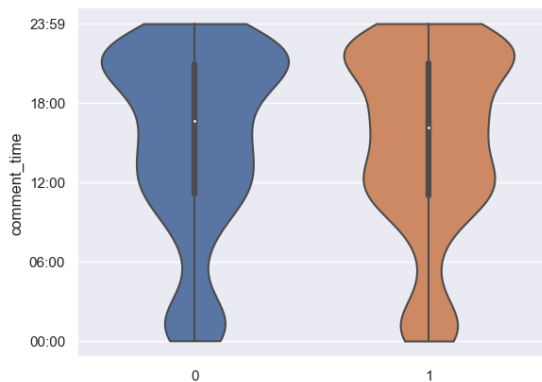


Fig. 10. Time of review.

Looking at the plots, we found the count of Chinese characters, English words, numbers and movie names to be effective features. The sentiment score, review score and time of review also provide profound insights into spam review detection. The

count of Chinese characters after data cleaning and the count of emojis, nonetheless, are removed from modelling due to dependence and ineffectiveness respectively.

4.5 Modelling

The training and testing accuracy for different models are summarised in Table II. The ROC curves and other metrics are also provided as reference in Fig. 12 and Fig. 13.

Table II
ACCURACY FOR MODELS

Classification algorithm	Training accuracy	Testing accuracy
KNN ($k = 4$)	86.17%	78.76%
Naïve Bayes	72.11%	65.31%
Logistic Regression	88.13%	81.88%
SVM with RBF kernel	82.97%	82.50%
SVM with linear kernel	82.19%	80.31%
Decision Tree	87.34%	81.25%

Naïve Bayes is significantly underperforming among all models. The problem of underfitting is severe amid various tuning attempts. This can be attributed to the fact that it is a very simple model which makes a very strong assumption on the independence of features. Once this assumption is not satisfied, the inflexible model becomes biased, thus could not be put to use in a real-world setting.

The performance of KNN is also less than ideal. This may be due to the curse of dimensionality as reviews may seem very close to each other in high dimensional space created by TF-IDF even when they are not actually similar to each other. Besides, strongly dependent to the value of K , KNN is very sensitive to any changes in this value. Although our chosen value of K is optimal in this dataset, it may not be generalised to other data. It would be impractical to retuning the value of K repeatedly in a real-world spam filter, rendering KNN unusable.

Logistic regression gives a satisfactory performance as its accuracy only slightly trails behind SVM. In practice, it is also easy to implement, interpret, and very efficient to train. Yet, we should bear in mind that performance may drop if the number of samples is inadequate.

Regardless of the kernel used, SVM in general is on par with, if not outperforms other models. The reason is that SVM auto-regularise while picking the thickest margin. Amid the high dimension of TF-IDF features, the use of RBF kernel did not give a notable increase in accuracy. Yet, the training time used for optimisation is much longer than that of a linear model, as the former takes 13.5 seconds while the latter takes merely 9.1 seconds. Despite the good performances, we should acknowledge that SVM does not perform well with large training sets because of the large training complexity, which is a crucial consideration when put into practical use.

Like logistic regression and SVM, decision tree performs well despite overfitting. Decision tree is prone to overfitting by default, although easy to construct and extremely fast during operation. With rigorous pruning techniques and limiting the maximum depth, we could alleviate some but not all overfitting in the model. In a practical setting, we suggest using random forest instead of a standalone decision tree so as to reduce the variance of prediction. Its high interpretability and visibility, as illustrated by the visualisation of our decision tree in Fig. 11, is also desirable in certain applications.

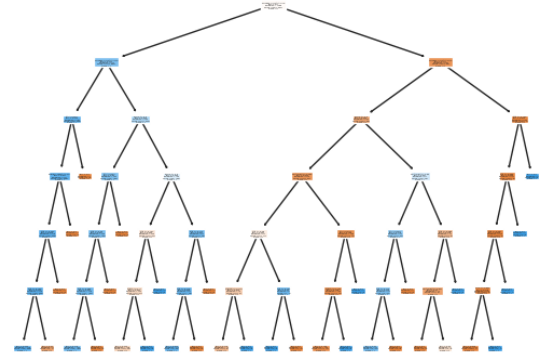


Fig. 11. Visualisation of the decision tree.

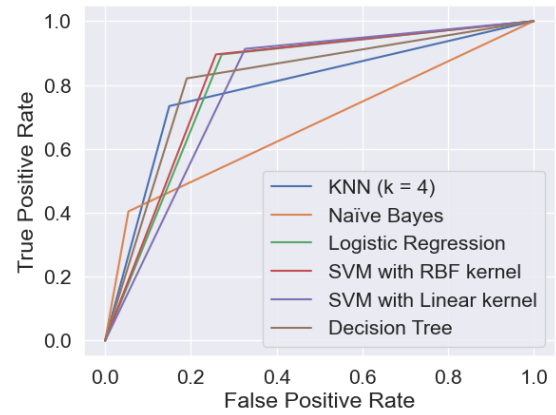


Fig. 12. ROC of the classifiers.

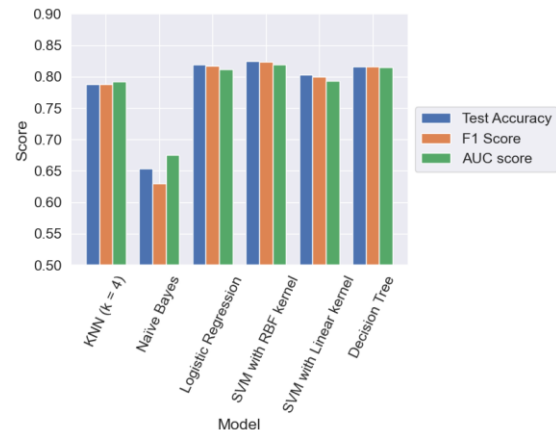


Fig. 13. Visualization of the classification results

5. Conclusion and Future Work

Fruitful conclusions can be drawn from the experimental results. Regarding the features, the useful features found are:

- Count of Chinese characters
- Count of English words
- Count of numbers
- Count of movie names
- Sentiment score of review
- Rating associated with the review
- Review posting time

In the process of applying various models spam review detection, we found that logistic regression, SVM and decision tree are the most effective. When building a spam filter in the real-world settings, the key consideration would be the sample size given in training. If the sample size is large, then logistic regression and decision tree are more desirable; otherwise, SVM should perform the best.

For future work, we may define more words into the dictionary of tokenisation by ourselves. For example, we may include the list of trending slangs, movie names and Chinese name of actors and actresses. We may also train our own sentiment analyser which can better understand the sentiment in the movie reviews using Chinese slang words. Moreover, more lexical features, such as the ease of reading, the complexity of the sentence composition and the number of Chinese words with respect to different part of speech using POS tagging technique, can be added. Furthermore, Linguistic Inquiry and Word Count (LIWC) tools may be employed to analysis the characteristic features of the language usage in the movie review.

It is also suggested that the movie reviews dataset should be enlarged by scraping more movies from a larger variety. Being the first of its kind, the Chinese labelled movie review dataset should be made available for the use of other researchers and public maintenance.

As neural networks grow mature in various applications, we could also apply it to the problem of spam movie review detection in the future, as it has been applied to spam commercial reviews [18]. Instead of focusing on feature engineering, much emphasis is attached to embeddings which are essential to mapping high-dimensional sentences into a low-dimensional embedding space. Convolutional neural network (CNN) and recursive neural network (RNN) could then be used to model semantic representations of sentences.

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